

Jayden Lieu

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Education

University of British Columbia – Faculty of Science (2024 - Present) – Data Science Major

Dean's List (2024-2025)

- DSCI 200 (Data Acquisition, Exploration, and Management)
- STAT 301 (Statistical Modeling)
- DSCI 221 (Data Structures & Algorithms)
- DSCI 320 (Data Visualization)

Work Experience

University of British Columbia – Undergraduate Teaching Assistant (September 2026 - Present)

CPSC 203 (Programming, Problem Solving, and Algorithms)

- Debugged, maintained, and automated python auto grader scripts to ensure precise code evaluation and test accuracy for student submissions
- Lead interactive tutorials for 40+ students, reinforcing core object-oriented programming principles, software logic, and debugging alongside assisting programming work such as sudoku solvers using Dijkstra's algorithm

Playland – Candy Lead (July 2025 - Present) | Playland – Candy Attendant (May 2022 - July 2025)

- Ensure quality control, inventory level accuracy, and manage production of food
- Supervised and trained over 20+ team members on operating procedures and customer service skills

Projects

NASA EONET Data Extraction & Privacy Analysis | R (February 2026 - April 2026)

- Gathered, cleaned, and structured nested JSON spatial coordinate data using scripts to build tidy datasets
- Implement data transformation, data collection, and quality assurance workflows to handle missing values, validate geospatial coordinate formatting, and evaluate privacy metrics to protect sensitive user records

Genomic Pipeline & Interval Tree | Python (February 2026 - March 2026)

- Implemented an interval tree search algorithm with several helper functions to query overlapping genomic features across 4M+ genomic intervals
- Optimized time complexity for querying genomic coordinates, enabling fast retrieval of positional gene data

PlaiCraft Dataset Project | R (September 2024 - December 2024)

- Identified a demographic of players for a research group to recruit in their project by analyzing game data
- Imported, wrangled and tidied data into a suitable form for data analysis using the dplyr, tidyr, and readr libraries, followed by exploratory analysis with distributions and scatterplots using ggplot2
- Trained a predictive KNN regression model, leveraging cross-validation to accurately analyze player behaviour trends such as age while preventing data bias

Technical Skills

Languages: Python (Pandas, NumPy, matplotlib, Vega-Altair), R (dplyr, tidyr, readr, reapr, ggplot2), SQL, HTML, PHP

Tools & Concepts: Git, Jupyter Notebook, Power BI, VS Code, Data QA/QC, Data Pipelines